

Python Fundamentals – Assignment

Total Questions: 30

Section A – One-Line Answer Questions

- Q1. What is Python?
- Q2. What is a variable in Python? Give one example.
- Q3. Name any four built-in data types in Python.
- Q4. What is the difference between int and float?
- Q5. What is the use of the type() function?
- Q6. What is the difference between = and ==?
- Q7. Name the arithmetic operators available in Python.
- Q8. What is type conversion? Give one example.
- Q9. What is the purpose of an if-else statement?
- Q10. What is the difference between a for loop and a while loop?

Section B – Describe with Example

- Q11. Explain Python variables with an example. Also mention how multiple variables can be assigned values.
- Q12. Explain the difference between mutable and immutable data types with examples.
- Q13. Explain arithmetic, relational, and logical operators with suitable examples.
- Q14. Explain implicit and explicit type conversion in Python with examples.
- Q15. Explain the if-elif-else structure with a suitable example.

Q16. Explain the for loop in Python. Write a program to print numbers from 1 to 10.

Q17. Explain a nested for loop with an example.

Q18. Explain the while loop with an example. When is it useful?

Q19. Explain the break statement with an example.

Q20. Explain the difference between List, Tuple, Set, and Dictionary with one example of each.

Section C – Programming Problems

Q21. Student Marks

Write a Python program to accept marks of a student and display:

A for marks 90 and above

B for marks 75–89

C for marks 60–74

D for marks 40–59

Fail for marks below 40

Also display Invalid Marks if marks are below 0 or above 100.

Q22. Write a Python program to accept 10 numbers from the user using a for loop and calculate:

- How many numbers are positive
- How many numbers are negative
- How many numbers are zero
- Sum of all positive numbers
- Sum of all negative numbers

At the end, display all five results.

Positive numbers:

Negative numbers:

Zeros:

Positive Sum:

Negative Sum:

Condition: Do not use a List to store the numbers. Use variables and a loop to solve the problem.

Q. 23 Student Performance Analyzer

Write a Python program to accept marks of 10 students using a for loop.

For each student, accept marks for:

Python

Java

SQL

Calculate the total and percentage for each student.

Based on the percentage, assign the result:

90 and above → Excellent

75-89 → Very Good

60-74 → Good

40-59 → Average

Below 40 → Fail

The program should finally display:

Total and percentage of each student

Number of students who passed

Number of students who failed

Highest percentage

Lowest percentage

Number of students scoring 75 or above

Challenge: Do not use List, Dictionary, Set, or Tuple. Use only variables, `for` loop, `if-elif-else`, operators, and type conversion.

Q24. Sum of Numbers

Write a Python program using a while loop to accept a number n and calculate the sum of numbers from 1 to n.

Example:

Input: 5

Output: 15

Q25. Multiplication Table

Write a program to accept a number from the user and print its multiplication table from 1 to 10.

Example:

Enter number: 7

7 x 1 = 7

7 x 2 = 14

...

7 x 10 = 70

Q26. Number Pattern

Write a Python program using a nested for loop to print the following pattern:

```
*
* *
* * *
* * * *
* * * * *
```

Q27. Stop When Number is Found

Write a program that repeatedly accepts numbers from the user.

If the user enters a positive number, continue.

If the user enters 0, stop the program using break.
Display the numbers entered before stopping.

Q28. List Operations

Create a list containing 10 student marks.

Write a program to:

Display all marks.
Display the highest mark.
Display the lowest mark.
Calculate the total marks.
Calculate the average marks.
Count how many students scored more than 75.

Q29. Dictionary – Student Information

Create a dictionary containing:

```
student = {  
    "name": "Rahul",  
    "age": 21,  
    "marks": 85,  
    "course": "Python"  
}
```

Write a program to:

Display the student's name.
Display the marks.
Update the marks.
Add a new key called city.
Display all keys.
Display all values.

Q30. Collection-Based Problem

Create a list containing duplicate numbers:

```
numbers = [10, 20, 10, 30, 20, 40, 50, 30, 60]
```

Write a Python program to:

Remove duplicate values using a Set.

Convert the Set back into a List.

Sort the resulting List.

Convert the List into a Tuple.

Display the final Tuple.

Create a Dictionary containing the number as the key and its square as the value.

Expected dictionary format:

```
{  
    10: 100,  
    20: 400,  
    30: 900  
}
```

Note:

Submission Guidelines

1. Write clean and properly indented Python code.
2. Use meaningful variable names.
3. Add comments wherever required.
4. Test each program with different inputs.
5. Write all theory questions with answers in Assignment notebook